1 A four-root elimination in twelve unrestricted integers

Result and comparison. The literal equation `integer12_degree512.json` has twelve independent integer unknowns, exact total degree 512, and 132 live binary operations (82 multiplications, 25 additions, 25 subtractions). Squares and scalar multiplications are charged. This is a new witness-elimination proof and a fixed-twelve-variable tradeoff, not a variable-count or degree record. Against the catalogued 12/500/139 point it saves seven operations at twelve more degrees; against 12/1084/123 it saves 572 degrees at nine more operations. The catalogue already contains the eleven-integer point 11/398/121, which dominates it across all three metrics. The retained 13/152/119 remains a separate lower-degree tradeoff. All triples here are variables/degree/operations.

The ordinary signed kernel and its native/compiler/syntactic-refutation calibration are inherited explicitly, not reproved by a finite checker. Composing that calibration with the total zero-set maps below proves in ordinary ZFC that this polynomial's universal integer nonvanishing is equivalent to syntactic Con(ZFC). No consistency assumption, consistency proof, new end-to-end proof-assistant certificate or global minimum is claimed.

1.1 Complete polynomial and tuple

The unrestricted variables, in literal order, are

$$j, C, z, h, r, v, w, T, c, n, p_1, p_2 \in \mathbb{Z}.$$

The six complete fixed integers are $\beta, K_0, M_C, M_F, A_{\text{cone}}, B_{\text{cone}}$, where M_F is already shifted, $\beta = 2^{1064921} - 1$, $A_{\text{cone}} = 2^{1113}$ and $B_{\text{cone}} = 2^{666104}$. All six byte payloads and their finite construction are supplied under `inherited_round20/inherited/coefficients`; no coefficient port is left unspecified. Define polynomial abbreviations

$$\begin{gathered}
J = 6j + 1, \quad q_- = \beta J, \quad q = q_- + 1, \\
W = 12w + 4, \quad X = qW, \quad Y = v, \\
F = (K_0 + W)C - zq_-, \quad R = [q(q - F) - C](q^2 - 1) + (M_C + qM_F)J, \\
E = XY, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\
\ell = EY, \quad k = R + 1 + h\ell, \quad \eta = c - Yk, \quad L = \ell k, \\
I = (2r + 1)c^2, \quad D_s = \Delta I^2, \quad F_s = 1 + D_s, \quad Q = \Delta D_s, \\
V = F_s(cT - R) - c, \quad q_3 = q^3, \quad M = 2q_3, \quad U = MH, \\
K_{\text{lo}} = X(2X - 3c) - 2, \quad D_{\text{err}} = (A_{\text{cone}}F - q)^2 + (B_{\text{cone}}C - q)^2, \\
B_0 = (Yk)^2(q^2 - D_{\text{err}}) - q^2[4(Y\eta)^2 + (kq^2)^2], \quad G = 4(B_0 - p_1^2 - p_2^2) - 3, \\
D_t = 1 + L(L + k), \quad U_2 = U^2, \quad Z_c = U(n + 2L) + MK_{\text{lo}} + H(Y - q_3), \\
\delta = Q - 1, \quad C_y = QV^2 - 1, \quad D_U = \delta U_2.
\end{gathered}$$

The five inputs to a homogeneous norm are

$$S = \delta Z_c^2, \qquad A = D_U F_s, \qquad B = 9D_U D_t, \qquad C_* = 729 D_U G, \qquad D = 81 U_2 C_y.$$

The following schedule defines the entire output:

$$\begin{gathered}
s_+ = S + D, \quad a_+ = A + C_*, \quad a_\times = AC_*, \\
e_2 = a_\times + Ba_+, \quad m = s_+ - a_+ - B, \quad m_2 = m^2, \\
d_2 = SD, \quad h_2 = m_2 + 4(d_2 - e_2), \quad e_3 = a_\times B, \\
n_0 = h_2^2 + 16(d_2 m_2 - 4e_3 s_+), \quad \boxed{P = n_0^2 - 64 d_2 (m h_2 - 16 e_3)^2.}
\end{gathered}$$

Every displayed register is an abbreviation. The literal DAG uses only integer leaves, addition, subtraction and multiplication.

1.2 The four-root identity and denominator clearing

Let $\mathcal{H}_4(S, A, B, C_*, D)$ denote the final schedule. It is homogeneous of degree eight in its five arguments and satisfies the formal identity

$$\mathcal{H}_4(z^2, A, B, C_*, D) = \prod_{\epsilon \in \{-1,1\}^4} (z - \epsilon_1\sqrt{A} - \epsilon_2\sqrt{B} - \epsilon_3\sqrt{C_*} - \epsilon_4\sqrt{D}).$$

This holds for arbitrary complex roots, including coincident square classes. The independent standard-library checker multiplies all sixteen factors exactly and compares all 495 coefficients.

Unconditionally, q is even and nonzero, $|X| \geq 8$, and $a = (X+1)Y$ is zero or has absolute value at least seven. Thus $\Delta = (a+2)^2 - 1 \geq 3$, H is odd and nonzero, and $U \neq 0$. Moreover $D_t, F_s \geq 1$ because $D_t = 1 + \ell(\ell+1)k^2$ and ℓ is even. As $Q = (\Delta I)^2$, either $Q = 0$ or $Q \geq 9$; hence δ, C_y never vanish. Also $G \equiv 1 \pmod 4$, so $G \neq 0$. With $\rho = Z_c/U$, the exact proof identity is

$$P = \delta^8 U^{16} \prod_{\epsilon \in \{-1,1\}^4} \left(\rho - \epsilon_1 \sqrt{F_s} - 3\epsilon_2 \sqrt{D_t} - 27\epsilon_3 \sqrt{G} - 9\epsilon_4 \sqrt{C_y/\delta} \right).$$

The factor δ^8 is already cleared in the literal polynomial; the quotient above is proof notation, not a division gate.

1.3 Separation and rational Pell integrality

The authentic β is $1 \pmod 3$. For every integer assignment,

$$q = X = -1, \quad W = 1, \quad a = H = \Delta = 0 \pmod 3,$$

so $F_s = D_t = 1$ and $\delta = C_y = -1 \pmod 3$. For D_t, use $\ell = -Y^2$ and $Y^4 - Y^2 = 0 \pmod 3$. In the order

$$\sqrt{F_s}, \quad 3\sqrt{D_t}, \quad 9\sqrt{C_y/\delta}, \quad 27\sqrt{G},$$

the squared 3-adic valuations are $0, 2, 4, 6 + v_3(G) \geq 6$.

Write each nonzero rational radicand as du^2 with d a signed squarefree integer and $u \in \mathbb{Q}^\times$. Distinct $\sqrt{d}$ are linearly independent over $\mathbb{Q}$, by sign-change character projections in a multiquadratic extension (including $\sqrt{-1}$). Consider only an irrational class $d \neq 1$; the rational carrier belongs to class 1. Any class meeting the first three roots has $v_3(d) = 0$, and its coefficients have a unique least valuation among $0, 1, 2, \geq 3$. Their signed sum cannot vanish. A class containing only the guard cannot vanish either. Consequently every root in a vanishing norm factor is rational. This includes negative radicands and dependent square classes; it does not assume that at most one term is imaginary.

At a zero there are signed rationals f, τ, t, g such that

$$f^2 = F_s, \quad \tau^2 = D_t, \quad t^2 = C_y/\delta, \quad g^2 = G, \qquad \rho = f + 3\tau + 9t + 27g.$$

The first, second and fourth are integers, and g is odd. For the third, $QV^2 - (Q-1)t^2 = 1$. If $Q = 0$, then $t = \pm 1$. If $Q > 1$, rational Pell integrality follows by descent. Indeed, write $t = r/m$ in lowest terms; $m^2 \mid Q - 1$. After sign reflection, $u = |V| > 1$ implies $u < t < Qu/(Q-1)$. The norm-preserving map

$$u' = (2Q-1)u - 2(Q-1)t, \qquad t' = (2Q-1)t - 2Qu$$

has $u' \in \mathbb{Z} \setminus \{0\}$, $|u'| < u$, and retains the reduced denominator m because $2Q - 1 \equiv 1 \pmod m$. Descent reaches $u = 1, t = 1$, forcing $m = 1$. Thus all four extracted witnesses are integers.

1.4 Total semantic maps

In the inherited fifteen-variable kernel put $Z_o = Un_o + MK_{\mathrm{lo}} + H(Y - q_3)$. Its residuals are

$$(Z_o + Uk)^2 + 4U(LZ_o - U) = 0, \qquad f_o^2 - \Delta I^2 - 1 = 0,$$
$$V_o^2 - (Q_o - 1)y_o(2V_o + y_o) - 1 = 0, \qquad B_0 - 1 - p_1^2 - p_2^2 - p_3(p_3+1) = 0,$$

where $Q_o = \Delta(f_o^2 - 1)$ and $V_o = f_o^2(cT - R) - c$. From every new zero restore

$$f_o = f, \quad y_o = t - V, \quad p_3 = (g-1)/2, \quad n_o = n - k - f - \tau - 9t - 27g.$$

Then $Z_o/U + 2L + k = 2\tau$, and

$$(Z + Uk)^2 + 4U(LZ - U) = (Z + U(2L + k))^2 - 4U^2 D_t$$

proves the first residual. The remaining three follow from the extracted square and Pell equations. Conversely, every old zero gives the integer

$$\tau = \tfrac{1}{2}(Z_o/U + 2L + k)$$

(the rational square is $4D_t$), together with $g = 2p_3 + 1$ and $t = V_o + y_o$. Set $n = n_o + k + f_o + \tau + 9t + 27g$ and delete f_o, y_o, p_3. Then a factor of the new norm vanishes. No canonical-witness restriction or unproved quotient integrality is introduced.

1.5 Exact degree and reproducibility

Let

$$q_0 = 6\beta j, \quad a_0 = 12 q_0 w v, \quad F_0 = 12 w C - q_0 z, \quad R_0 = -q_0^3 F_0, \quad U_0 = 8 q_0^3 a_0.$$

Also $Q_0 = 4a_0^4 r^2 c^4$ and $V_0 = -4a_0^2 r^2 c^4 R_0$. The degrees of (S, A, B, C_*, D) are bounded by $(48, 42, 48, 46, 64)$. The last is attained, with entire leading form

$$D_0 = 81 U_0^2 Q_0 V_0^2 = 2^{12} 3^4 q_0^6 a_0^{10} r^6 c^{12} R_0^2.$$

Homogeneity and the coefficient $[D^8]\mathcal{H}_4 = 1$ show that $[P]_{512} = D_0^8$. Every other norm monomial has strictly smaller weighted degree. In particular

$$[j^{192} w^{80} v^{80} r^{48} c^{96} z^{16}] P = 2^{448} 3^{304} \beta^{192} > 0.$$

This authentic fixed-coefficient witness has total degree 512.

The portable commands `build.py --check`, `check.py`, and `reconstruct_coefficients.py` require only the Python standard library. Normal and optimized runs authenticate all twelve live variables, all six coefficient ports, the complete 495-term norm, the restore identities, the leading form, 729 modulus-three fixtures, and rejection of all 132 live-gate corruptions. All six full numerals (615,883 raw bytes) are rebuilt from the retained finite recipe. Source pins distinguish current comparison commit `892d73e1ab4bc5fd34ecca1967b06dc47ca95494` from the immediate round20 parent and its source153 signed packet. Finite checks do not replace the unbounded separation/descent proof or inherited compiler calibration.